λ HAOBIN (HIROKI) CHEN λ

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EDUCATION

Indiana University Bloomington, IN, USA *2023 - Dec. 2027 (Expected)*
Ph.D. in Computer Science
Advisors: Haixu Tang, Chenghong Wang (IU) & XiaoFeng Wang (Nanyang Technological University)

Nankai University, Tianjin, China *2019 - 2023*
B.Eng. in Information Security

RESEARCH INTERESTS

Confidential Computing, Systems Security, and Formal Verification.

INDUSTRIAL EXPERIENCE

Ant Group May 2026 - Aug. 2026
Research Intern *Beijing, China*

- Designed and implemented RCU and weak memory extension to the codebase [11 kLoc].
- Implemented the concurrent verification for MM subsystem based on the Iris-like resource algebra.

CertiK Inc. Oct. 2025 - Mar. 2026
Security Research Intern (Mentors: Dr. Hao Chen, Dr. Sean N. Anderson) *New York, NY*

- Applied **Verus** to verify the memory management module of the **Asterinas** Rust-based OS kernel (ATC'25), ensuring spatial and temporal memory safety.
- Conducted formal verification of Solana smart contracts and protocols to provide mathematical assurance for digital assets.

Privacy Innovation Lab, TikTok Inc. May 2024 - Aug. 2024
Research Intern (Mentor: Dr. Mingshen Sun) *San Jose, CA*

- Proposed **TAVERNS**, a decentralized remote attestation paradigm that eliminates reliance on centralized verification services.
- Developed a TEE-based verification solution using **Zero-Knowledge Proofs (ZKP)**, resulting in a **U.S. Patent application**.
- Engineered data clean rooms (ManaTEE, part of the *Confidential Computing Consortium*) for TikTok's research platform on Google Cloud TEE instances using k8s.

Google Summer of Code: Apache Teaclave Jun. 2023 - Nov. 2023
Open Source Contributor *Remote*

- Developed data analysis solutions for privacy policy enforcement using Trusted Execution Environments.
- Integrated formal verification components to enhance the framework's security guarantees.

ACADEMIC EXPERIENCE

Verifiable Policy Enforcement in Confidential VMs Jul. 2026 - Present
Research Assistant (with Prof. Chenghong Wang, Mingshen Sun & Benjamin S. Ternthal) *IU*

- Designed intra-CVM isolation to protect trusted governance logic from an untrusted guest OS.
- Built a Verus-verified IFC monitor for policy enforcement and provenance tracking.
- Bound provenance and policy events to hardware attestation for cloud-native workloads.

Privacy-Preserving Data Sharing and Analytics (PPDSA) Aug. 2024 - Present
Research Assistant (with Prof. XiaoFeng Wang, Chenghong Wang (IU), Marcela Milara (Intel) *IU*

- Applied state-of-the-art confidential computing technology to securing data analytics at large scale.
- Applied rigorous machine-assisted verification and policy enforcement for data lakehouses.
- Explored hardware-software co-designs for modern confidential data analytics.

Center for Distributed Confidential Computing (CDCC) Aug. 2023 - Present
Research Assistant (with Prof. Haixu Tang & XiaoFeng Wang) *IU*

- Architected hardware-software co-designs for policy-enforced isolation within TEEs.
- Utilized **Verus** and **Coq** to verify large-scale codebases and security-critical system kernels.
- Optimized secure system performance for CPU/GPU TEEs in cloud and edge environments.

Proof of Being Forgotten: Rust-SGX Enclave Framework May 2022 - Jun. 2023
Research Assistant (with Prof. XiaoFeng Wang & Dr. Mingshen Sun) *Remote*

- Implemented secret residue cleaning algorithms and custom allocators for Intel SGX using **Rust**.
- Formally verified execution models in **Coq** to ensure no secret leakage to unauthorized parties.

PUBLICATIONS

- **Haobin Hiroki Chen** *et al.* [redacted]. In submission.
- Hao Wang, **Haobin Hiroki Chen**, Hongbo Chen, Weijie Huang, Genevieve Mortensen, Yingjie Chen, Jing Su, Haixu Tang. A Confidential Computing Framework for Runtime Data Policy Enforcement in Biomedical Data Analytics: Design, Implementation, and Evaluation. To appear in the *60th Hawaii International Conference on System Sciences (HICSS)*, 2027.
- **Ant Research**. Towards Verification of Weak Memory Model and the RCU Module in the Asterinas Kernel. 2026. **available soon**
- Y. Guo, H. Chen, **Haobin Hiroki Chen**, Y. Luo, X. Wang, C. Wang. BOLT: Bandwidth-Optimized Lightning-Fast Oblivious Map powered by Secure HBM Accelerators. In *ACM CCS* (**CCS'25**), 2025.
- **Haobin Hiroki Chen**, H. Chen, M. Sun, C. Wang, X. Wang. PICACHV: Formally Verified Data Use Policy Enforcement for Secure Data Analytics. In *USENIX Security* (**Sec'25**), 2025.
- **Haobin Chen**, Y. Yang, S. Lv. Revisiting frequency-smoothing encryption: new security definitions and efficient construction. *Cybersecurity* (7), 15 (2024).
- H. Chen, **Haobin Hiroki Chen**, M. Sun, K. Li, Z. Chen, X. Wang. A Verified Confidential Computing as a Service Framework. In *USENIX Security* (**Sec'23**), 2023.

TECHNICAL SKILLS

Programming	Rust, C/C++, Python, OCaml, Haskell
Verification	Verus, Coq/Rocq, Lean4

HONORS AND AWARDS

2026 **Research Funds from the Confidential Computing Consortium**

2024 **ACM CCS Distinguished Artifact Reviewer**

2023 **Nankai Distinguished Bachelor Thesis Award** (Top 3%)

2021 **3rd Prize, National College Student Information Security Contest** (Top 8%)

2021/22 **Nankai Academically Excellent Student Scholarship** (Top 5%)

ACADEMIC SERVICES

Program Committee (Artifact Evaluation): ACM CCS ’24, USENIX Security ’25, NDSS ’26 ’27
External Reviewer: IEEE TIFS, ACM SIGMOD ’26, PeerJ Computer Science